

Full Stack Development

Lab Exercise

1. Source Code

```
```html
<!DOCTYPE html>
<html lang="en">
<head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 <title>SOC-AI | Cybersecurity Scanner</title>

 <script src="https://cdn.tailwindcss.com"></script>

 <script>
 tailwind.config = {
 theme: {
 extend: {
 colors: {
 neon: "#39ff14",
 dark: "#050805",
 panel: "#0a110a"
 },
 fontFamily: {
 mono: ["Courier New", "monospace"]
 },
 boxShadow: {
 glow: "0 0 25px rgba(57, 255, 20, 0.2)"
 }
 }
 }
 }
 </script>

 <style>
 body {
 background-color: #050805;
 background-image:
 linear-gradient(rgba(57, 255, 20, 0.03) 1px, transparent
1px),
 linear-gradient(90deg, rgba(57, 255, 20, 0.03) 1px,
transparent 1px);

```

```

 background-size: 40px 40px;
 }

 .glow {
 text-shadow: 0 0 12px rgba(57, 255, 20, 0.8);
 }

 .terminal-line::before {
 content: "> ";
 color: #39ff14;
 }

 .scan {
 animation: blink 1.2s infinite;
 }

 @keyframes blink {
 50% {
 opacity: 0.4;
 }
 }
</style>
</head>

<body class="text-gray-300 font-mono">

 <header class="border-b border-green-900 bg-black/90 sticky top-0 z-50">
 <div class="max-w-7xl mx-auto px-6 py-5 flex flex-col md:flex-row
items-center justify-between gap-5">

 <div class="flex items-center gap-3">
 <div class="w-10 h-10 border border-neon flex items-center
justify-center text-neon font-bold shadow-glow">

 &
 </div>

 <div>
 <h1 class="text-xl font-bold text-white glow">SOC-AI</h1>
 <p class="text-xs text-green-700">SECURITY OPERATIONS //
AI</p>
 </div>
 </div>

 <nav>
 <ul class="flex flex-wrap justify-center gap-6 text-sm">

```

```

 <a href="#home" class="text-neon hover:text-white
transition">HOME

 <a href="#features" class="hover:text-neon
transition">FEATURES

 <a href="#scanner" class="hover:text-neon
transition">SCANNER

 <a href="#about" class="hover:text-neon
transition">ABOUT

</nav>

 <div class="hidden lg:flex items-center gap-2 text-xs
text-green-500">
 <span class="w-2 h-2 rounded-full bg-neon
shadow-glow">
 SYSTEM ONLINE
 </div>

</div>
</header>

<main>

 <section id="home" class="min-h-[90vh] flex items-center">
 <div class="max-w-7xl mx-auto px-6 py-20 w-full">

 <div class="grid lg:grid-cols-2 gap-16 items-center">

 <div>

 <div class="inline-flex items-center gap-2 border
border-green-900 bg-green-950/20 px-4 py-2 text-xs text-neon mb-8">
 <span class="w-2 h-2 bg-neon rounded-full
scan">

 THREAT DETECTION ENGINE ACTIVE
 </div>

 </div>

 </div>

 </div>

 </section>

</main>

</div>
```

```
<p class="text-green-700 text-sm mb-4 terminal-line">
 INITIALIZING SECURITY PROTOCOL
</p>

<h2 class="text-5xl md:text-7xl font-bold text-white
leading-tight mb-6">
 CODE
 SECURED.

 THREATS
 DETECTED.
</h2>

<p class="text-gray-500 max-w-xl text-base md:text-lg
leading-8 mb-8">
 SOC-AI analyzes your source code using artificial
intelligence
 and security analysis techniques to identify
potential
 vulnerabilities before they become a problem.
</p>

<div class="flex flex-col sm:flex-row gap-4">
 <a href="#scanner"
 class="bg-neon text-black px-7 py-3 font-bold
text-sm text-center hover:bg-white transition shadow-glow">
 [START SCAN]

 <a href="#features"
 class="border border-green-900 text-neon px-7
py-3 text-sm text-center hover:border-neon transition">
 VIEW FEATURES

</div>

</div>

<div class="border border-green-900 bg-black/80
shadow-glow">
 <div class="border-b border-green-900 px-5 py-3 flex
justify-between text-xs">
```

```

 soc_ai_terminal
 ● LIVE
 </div>

 <div class="p-6 space-y-4 text-sm">

 <p class="text-green-700">
 root@soc-ai:~$./security_scan
 </p>

 <p class="text-gray-500">
 Loading security engine...
 </p>

 <p class="text-gray-400">
 [OK] Code
analyzer loaded

 </p>

 <p class="text-gray-400">
 [OK] Threat
detection loaded

 </p>

 <p class="text-gray-400">
 [OK] AI model
loaded

 </p>

 <div class="border-t border-green-950
my-5"></div>

 <div class="grid grid-cols-2 gap-4">

 <div class="border border-green-950 p-4">
 <p class="text-gray-600 text-xs">FILES
SCANNED</p>

 <p class="text-2xl text-white
mt-2">128</p>

 </div>

 <div class="border border-green-950 p-4">
 <p class="text-gray-600
text-xs">THREATS</p>

```

```
<p class="text-2xl text-neon mt-2">03</p>
</div>

<div class="border border-green-950 p-4">
 <p class="text-gray-600 text-xs">RISK
LEVEL</p>

 <p class="text-2xl text-yellow-500
mt-2">MED</p>

</div>

<div class="border border-green-950 p-4">
 <p class="text-gray-600
text-xs">CONFIDENCE</p>

 <p class="text-2xl text-white
mt-2">94%</p>

</div>

</div>

<p class="text-green-700 pt-3">
 root@soc-ai:~$ <span class="text-neon
scan">_

</p>

</div>
</div>
</section>

<section id="features" class="border-t border-green-950 py-20">

 <div class="max-w-7xl mx-auto px-6">

 <div class="mb-12">
 <p class="text-neon text-sm mb-3">// SECURITY MODULES</p>
 <h2 class="text-3xl md:text-4xl text-white font-bold">
 Built to find what others miss.
 </h2>
 </div>

 <div class="grid md:grid-cols-3 gap-5">
```

```

<article class="border border-green-950 bg-panel p-7
hover:border-neon transition">
 <div class="text-neon text-2xl mb-6">01</div>
 <h3 class="text-white text-xl font-bold mb-4">
 Vulnerability Detection
 </h3>
 <p class="text-gray-500 leading-7 text-sm">
 Detect potentially dangerous patterns and common
 vulnerabilities hidden inside your source code.
 </p>
</article>
<article class="border border-green-950 bg-panel p-7
hover:border-neon transition">
 <div class="text-neon text-2xl mb-6">02</div>
 <h3 class="text-white text-xl font-bold mb-4">
 AI Code Analysis
 </h3>
 <p class="text-gray-500 leading-7 text-sm">
 Machine learning models analyze code patterns and
 identify suspicious or unusual behavior.
 </p>
</article>
<article class="border border-green-950 bg-panel p-7
hover:border-neon transition">
 <div class="text-neon text-2xl mb-6">03</div>
 <h3 class="text-white text-xl font-bold mb-4">
 Risk Assessment
 </h3>
 <p class="text-gray-500 leading-7 text-sm">
 Security findings are organized by severity to
 help
 developers understand which issues need
 attention.
 </p>
</article>
</div>
</div>
</section>
<section id="scanner" class="border-t border-green-950 py-20">

 <div class="max-w-5xl mx-auto px-6">

 <div class="border border-green-900 bg-black p-8 md:p-12
text-center shadow-glow">

```

```
<p class="text-neon text-sm mb-4">
 // SECURE PROJECT ANALYSIS
</p>

<h2 class="text-3xl md:text-5xl font-bold text-white
mb-5">
 Ready to scan your code?
</h2>

<p class="text-gray-500 max-w-2xl mx-auto leading-7
mb-8">
 Upload your project and let SOC-AI inspect your
source
 code for potential security vulnerabilities.
</p>

<div class="border border-dashed border-green-800 p-10
mb-7">

 <div class="text-4xl text-green-800 mb-4">
 [ZIP]
 </div>

 <p class="text-gray-400 text-sm mb-2">
 DROP YOUR PROJECT HERE
 </p>

 <p class="text-gray-700 text-xs">
 Supported format: .ZIP
 </p>

</div>

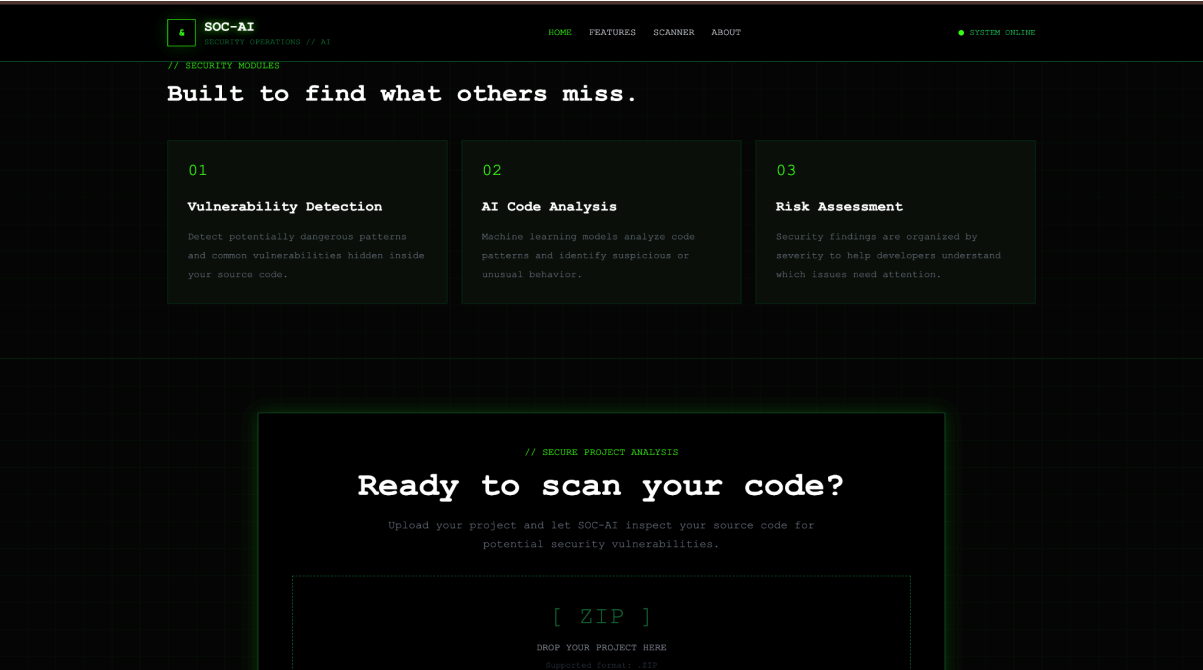
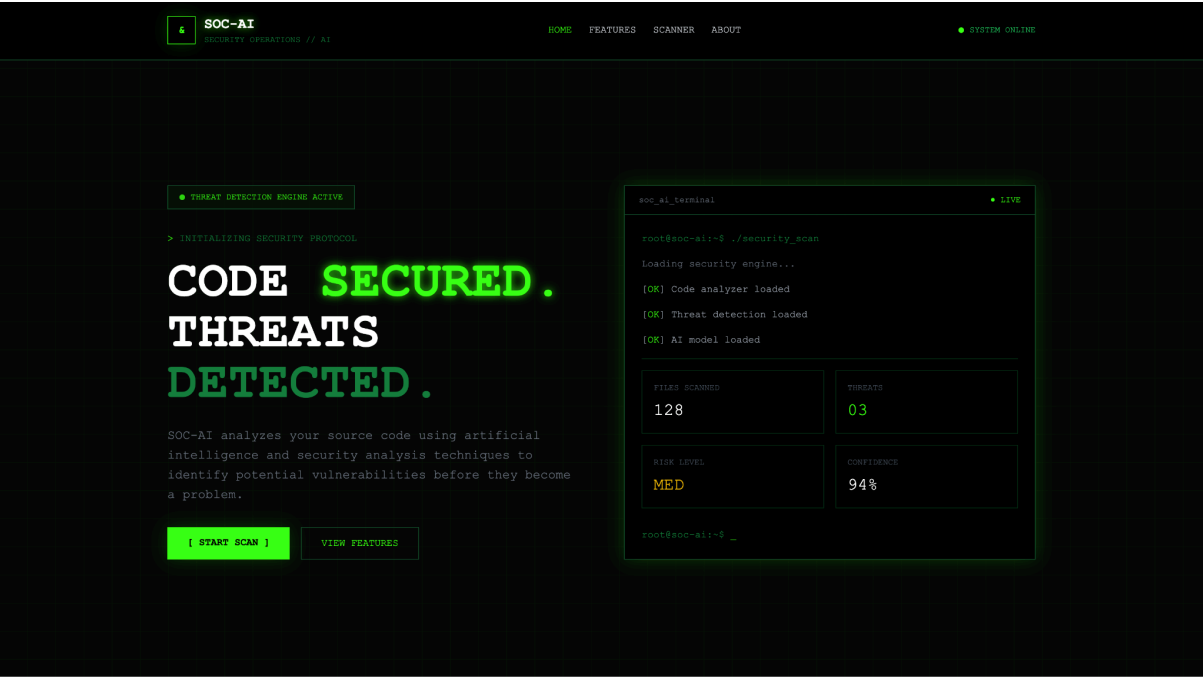
<button class="bg-neon text-black px-8 py-3 font-bold
hover:bg-white transition">
 UPLOAD & SCAN
</button>

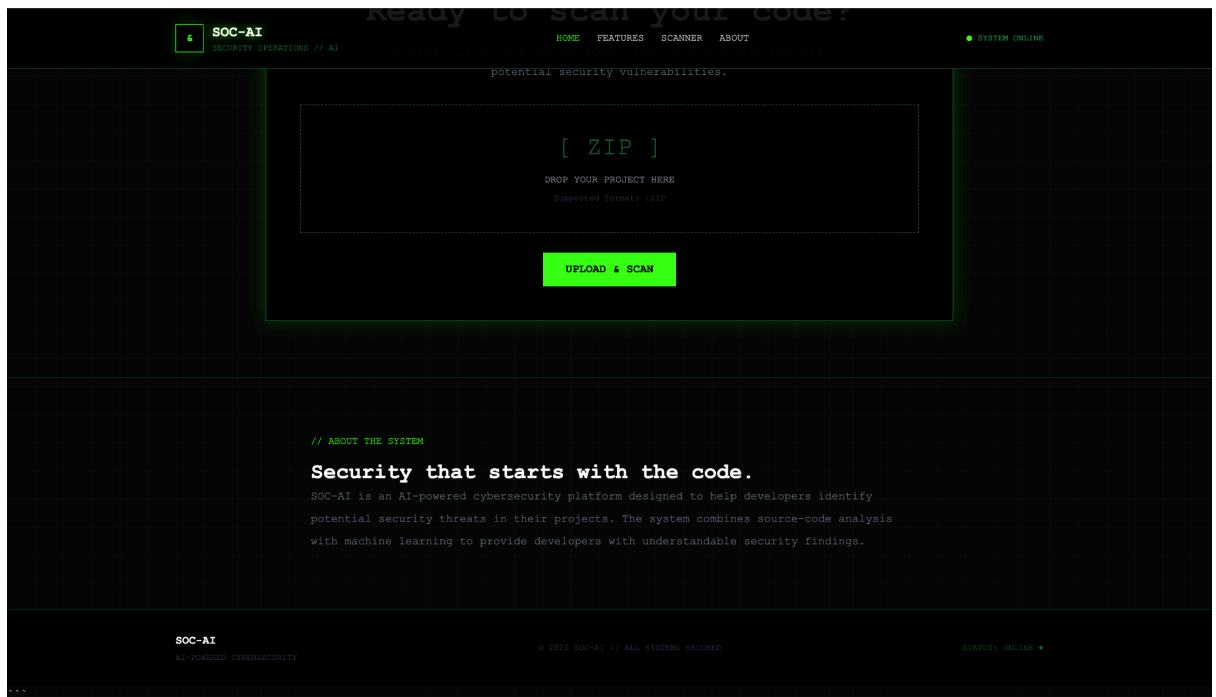
</div>

</div>
</section>
<section id="about" class="border-t border-green-950 py-20">
 <div class="max-w-4xl mx-auto px-6">
 <article class="prose prose-invert max-w-none">
```

```

 <p class="text-neon text-sm">// ABOUT THE SYSTEM</p>
 <h2 class="text-white text-3xl font-bold mt-4">
 Security that starts with the code.
 </h2>
 <p class="text-gray-500 leading-8">
 SOC-AI is an AI-powered cybersecurity platform
designed
 to help developers identify potential security
threats
 in their projects. The system combines source-code
 analysis with machine learning to provide developers
 with understandable security findings.
 </p>
 </article>
 </div>
</section>
</main>
<footer class="border-t border-green-950 bg-black">
 <div class="max-w-7xl mx-auto px-6 py-8">
 <div class="flex flex-col md:flex-row justify-between
items-center gap-4">
 <div>
 <p class="text-white font-bold">SOC-AI</p>
 <p class="text-gray-700 text-xs mt-1">
 AI-POWERED CYBERSECURITY
 </p>
 </div>
 <p class="text-gray-700 text-xs">
 © 2026 SOC-AI // ALL SYSTEMS SECURED
 </p>
 <p class="text-green-800 text-xs">
 STATUS: ONLINE ●
 </p>
 </div>
 </div>
</footer>
</body>
</html>
```





GitHub Link

<https://github.com/yashi-cpu/FSD-1.git>

I used ChatGPT as an AI assistance tool while developing this webpage. It was mainly used for generating ideas for the website layout, choosing suitable Tailwind CSS classes, improving the cybersecurity-themed design, and solving minor coding issues. I reviewed and modified the generated code according to the requirements of the exercise and my project idea.

This project is based on cybersecurity and aims to create a web-based platform for detecting security vulnerabilities in source code. The current webpage is designed as the frontend of the platform using HTML5 and Tailwind CSS. In the future, it can be connected with

a backend and AI/ML model to scan uploaded source code and identify possible security threats.